HELLO, MY NAME IS SMARTTRAM, HUMAN FACTORS IS ON BOARD, ENJOY THE RIDE! DEVELOPING A HUMAN FACTORS PROGRAM FOR AUTOMATIC TRAMS

Hanna Koskinen, Susanna Aromaa and Vladimir Goriachev
VTT Technical Research Centre of Finland Ltd
Tekniikantie 21, PL1000, 02044 VTT, Finland

ABSTRACT

Our transport system is currently undergoing fundamental change due to increasing use of automation. New automation solutions are introduced in all sectors of transportation, for example automated metros and autonomous ships are in the visions of technology developers. There are many reasons for this ongoing trend of higher use of automation such as demands for sustainability and efficiency to mention some. In this paper, we present a research and development effort aiming at introducing an automatic tram, that is to say SmartTram. In particular, we concentrate on how the changing role of human (as driver, passenger and member of other stakeholder groups) is acknowledged in the design of a new automatic tram. For this reason, we present a human factors engineering program for automated trams. This approach can be utilized also in other sectors when increasing automation.

KEYWORDS

Public Transport, Automatic Tram, Human Factors Engineering

1. INTRODUCTION

There is a worldwide initiative to increase use of automation solutions in transport systems and enable new kind of mobility services that are in the center of the smart city concepts. Even though the research and development of automated cars may have gain more attention (Cohen et al., 2017; Kyriakidis et al., 2019; Merat & Jamson, 2009) the same automatization trend impacts all other aspects of transportation (e.g., public transportation and cargo) (Beiker, 2014; Karvonen et al., 2011; Wahlström et al., 2016). In this paper, our focus is on public transportation and in particular developing automatic solutions for trams. The design task, that is to say, the development of automatic tram, is challenging because the tram needs to operate in an open infrastructure opposite to the closed-rail systems that most of the introduced automatic transport systems has been earlier. Moreover, apart from few exceptions, not many reference cases about automatic trams may be found (see e.g., Hofmann, 2020). The urban city environments that the trams often operate are, from the automation design point of view, the most challenging ones. In these urban environments, advanced sensor networks and high connectivity are demanded to reliable detect variety of objects (e.g., pedestrians and other vulnerable road users [VRU]) and aspects of the driving environment (e.g., weather and lighting conditions or traffic density).

From the operation and human point of view, the automatic tram will change the tram driver's work substantially. The tram driver will no longer be expected handling all the operative level driving tasks (e.g., driving by sight, controlling driving speed and stopping at tram stops as well as monitoring the situation in the cabin and the tram stops) and the main task shifts toward supervising the automatic driving system. Thus, even with the increasing levels of automation, the human operator is still needed at least to supervise and monitor that the system operates as intended. The driver may also be expected to intervene in the operation in case of an emergency or some complex driving situation (e.g., handling moving to detour or passing a traffic work site). From the passengers' point of view, the automatic tram is a notable change, as well. There may be issues such as can it be trusted in the functioning of the automatic tram (Karvonen et al.,

2011) or uncertainty how to behave and use the tram if there is no human driver to ask help. There are many studies that have been investigating the interaction of the pedestrians and other VRUs with the automatic vehicles (Hulse et al., 2018; Parkin et al., 2016; Rasouli & Tsotsos, 2018). In these studies, different visualization and signaling concepts have been presented and evaluated in order to communicate the intentions of the automatic vehicle to the other road users. In addition, there are also other stakeholders to consider during the design of automatic trams for example maintenance technicians. Due to several stakeholder/user groups with varying needs and expectations, driving a tram and the human interaction with the automated tram will remain as an important human factors issue to be taken under careful investigation.

In this paper, we introduce the human factors engineering (HFE) program including iterative evaluation of future solution that have been developed for the SmartTram project in order to acknowledge appropriately the human and social aspect in the design of the automatic tram. We first discuss the importance of developing proper HFE program in safety-critical design projects. After which we introduce the specific HFE activities planned for the SmartTram project. A particular emphasis has been put on laying out the verification and validation (V&V) plan and the related simulator and training facilities.

2. HUMAN FACTORS ENGINEERING IN SAFETY-CRITICAL DESIGN PROJECTS

A systematic approach and human factors practices are needed in the design of safety-critical systems such as control rooms of nuclear power plants, cockpits of aircrafts and operating theatres in hospitals where many technical but also social and organizational aspects need to be seamlessly integrated. In addition, maintaining holistic and life cycle perspective are also important in order to reach a proper socio-technical understanding of the design task at hand. It has been suggested in the literature, that at least the following list of HFE activities should be included in design of complex systems: screening, management and planning, operating experience (OE) review, development of concept of operations (ConOps), function allocation and analysis, task analysis, human reliability analysis, personnel selection, training and qualifications, system design and development, design of maintenance activities, V&V, and commissioning and in-time operation monitoring. All these HFE activities should be carried out in a coordinated manner and at the same time, there should be a continuous, effective and timely dialogue between the human factors representatives and the other engineering specialists working in the project. This is not a straightforward task to do. Consequently, the importance of HFE integration in complex engineering projects is widely recognized (Hale et al., 2007; Hugo, 2000) The benefits of HFE integration may be, for example, that the problems can be avoided or mitigated and, design relevant information and feedback from the evaluations can be more effectively fed back to the design. Thus, drafting a more formal HFE program within which all the HFE activities, their deliverables and main interactions with the other design parties in a project are described is one way to address the integration problem and create a base for high quality HFE practices. The HFE program should cover all the phases of the design process from the initial clarification and analysis phase to the final implementation of the system. Because of the different starting points and contextual circumstances in different design projects it is expected that the HFE program will need to be tailored and customized individually to each project even though the basic structure of the HFE program would be much alike. Based on these requirements, we present a HFE program for safety-critical system design and SmartTram (Table 1).

Table 1. A HFE program for SmartTram system design

Input	Design process	Output
• Need for change (e.g., growing amount of passengers, sustainable and environmentally friendly public transport solutions) • Existing concepts (e.g., proofs of concept, level of automation) • Operational experience (other automated traffic solutions e.g., metro)	**Initial clarification and analysis** Impact, needs and constraints HFE goals	• Human factors input to project plan • HFE program/ activities (i.e., tasks and responsible persons, schedule and resources) • Preliminary evaluation schema (i.e., Systems Usability Case based evaluation of concept, subsystems and integrated system)
• Operating environment description • User and safety requirements • Standards	**Requirements and concept design** Concept of operation Task analysis User requirements	• SmartTram concept (documentation of ConOps, level of automation, user interactions) • Project requirement set • HSI style guide (e.g., main design principles, HSI guidance on layout, navigation, color-coding, alarms etc.)
• I&C architectures • Standards	**Detail/integrated design** User interface, training design	• Tram interior and user interface design solution
• V&V plan (e.g., users groups, scenarios, acceptance criteria) • Standards	**Verification & validation (V&V)** Simulations Validation tests	• V&V record (e.g., identified human engineering discrepancies and design feedback, decisions on further design iterations)
	Implementation & commissioning	• Accumulated evaluation record (careful documentation of implantation and early operating experiences)

3. HUMAN FACTORS ENGINEERING IN SMARTTRAM PROJECT

The social and human aspect is important in the SmartTram project as the tram will operate in our everyday living environment and thus should be sensitive to how it is perceived also by the general public not to mention the tram drivers whose work is anyway going to change radically. For that reason a HFE program and a specific plan for the iterative evaluation of the SmartTram from human factors point of view was prepared and implemented as a part of the development process.

3.1 Human Factors Engineering Program

The HFE activities summarized in the Table 1 are the main responsibility of the HFE experts working in the SmartTram project. At the moment, the project is in its initial clarification and analysis phase, that is to say, a lot of analytical research work is carried out (e.g., options for technical platform are investigated.) The relevant information from the HFE perspective is also collected (e.g., identifying and analyzing the different user groups), and benchmarking on, for example, about the existing human-system interaction concepts in tram driving are conducted. In addition, operating experiences of similar systems are of interest in this phase. It may be that not many directly comparable solutions (i.e., automatic trams) can be found, however, operating experiences on other automatic transport and machinery may offer relevant background information as well. This information serves as an input for planning the SmartTram project and drafting the specified HFE program (i.e., planned HFE activities within the project) and preliminary evaluation schema (i.e., V&V plan).

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3.2 Verification and Validation Plan

Iterative evaluation and testing is an integral part of designing complex safety-critical systems (see Figure1) and the continuous engineering paradigm that promote improvements throughout the product life cycle. Thus, the HFE evaluation schema should be considered already from the very beginning of any development project. That is also why in the SmartTram project, the V&V plan was drafted in the first phase of the project (i.e., initial clarifications and analysis phase in the Table 1). Typical for complex system engineering projects, like the SmartTram-project, are that they are long and therefore the design work is often divided into several stages or manageable parts that follow each other's. A concept of subsystem validation (SSV) (Laarni et al., 2014) has been introduced to carry out evaluations throughout the design whenever a part of a whole get completed and ready for testing. For example, a modernization of a control room may be realized in stepwise and thus the evaluation of different subsystems (i.e., normal operation and safety systems) become timely in different time windows in the project. Following the SSVs, an integrated system validation (ISV) is conducted before the commissioning phase of the new system. It is also essential that when the system is in its operative state, the operating experiences are systematically collected, and used as an input for identifying further places of improvements for the system.

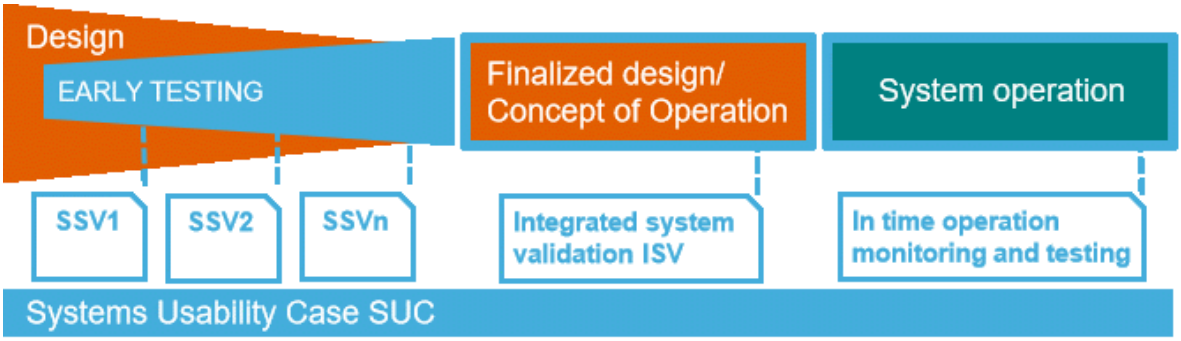


Figure 1. Iterative evaluation throughout the life cycle of a complex safety-critical system

Moreover, it is expected that a huge amount of HFE data being gathered even within one complex system engineering project and the life cycle of system. An evaluation approach called Systems Usability Case (SUC) (Koskinen et al., 2017; Liinasuo & Norros, 2007) has been used as a methodical starting point in the SmartTram project for managing and organizing the system evaluation. A safety case (Bishop & Bloomfield, 1998) is required in many safety standards and therefore produced in variety safety-critical domains as a part of their normal qualification procedure. SUC approach bases on Systems Usability (SU) construct (Savioja, 2014) and the safety case thinking and it exploits the design requirements in forming the set of criteria for the HFE evaluations. The SUC approach is considered to be especially well suited to evaluation of systems that are complex and unique by their nature. In the first part of the SUC procedure called as a goal structure, the project specific design requirements are organized thematically with regards to the more generic usability and HFE requirements and the high-level acceptance goal set for the project. Through this construction, a more specified set of acceptance criteria and the test occasions (e.g., scenarios) in which the fulfilment of the requirements may be tested are derived. Going through the design rational and the documentation helps also the HFE evaluators to become better acquainted with the system and thus be able to ask more specific and detailed questions concerning the systems usability. After constructing the goal structure, an evaluation session is realized to collect evidence. These evaluation sessions may be carried out with different type of mockups and prototypes or in more advanced simulator environments depending on the maturity of the design solution and the project phase. In the second part of the SUC, called as a claim structure, a reasoning process takes place. The claim structure comprises of a set of evidence, arguments and claims. Constructing the claim structure aims to create a documented body of evidence providing a valid argument of the degree of systems usability of the system under consideration. Often it is so that in the early phase evaluations many human engineering discrepancies (HEDs) are still identified. Each identified HED needs to be handled individually and give an appropriate resolution. Solving one HED may require redesign to take place whereas for other HED training of personnel may be the most effective way to handle the identified issue. When

coming closer to the implementation phase and the ISV of the system it is expected that the amount of HEDs is lower as many of the HFE issues are already been able to solve and act upon earlier in the V&V process. No critical HEDs should be existing in the system at the time of implementation. Using SUC helps to conduct the V&V process in a systematic, transparent and longitudinal fashion.

3.3 SmartTram Simulator Facility for User Testing

One useful approach to be applied during an HFE program in a safety-critical system design is virtual reality (VR) based models and simulators. In VR, digital technology provides simulation of reality typically through visual, auditory, and haptic interfaces. By using VR simulators in the context of safety-critical system design, it is possible to immerse users to their future operating tasks in a safe environment. The VR simulators exist realized with varying technical solutions (e.g., from visualizations on a PC screen to immersive virtual environments experienced with head mounted displays) and they can be used for different purposes during the HFE program. For example, by using a simulator it is possible to train new operators / tram drivers to cope hazard traffic situations. In general, there are several reasons contributing to VR simulators' usefulness, such as converting the abstract to the concrete (e.g., making some specific concept idea illustratively tangible) and making hands-on training possible in situations where it is still impossible to train things in the real world setting.

In the SmartTram project, a virtual environment was created for the SmarTram system (see Figure 2). A virtual prototype visualizes quite a conventional tram cockpit but it has been enhanced with new SmartTram features and control panels. The tram driver may now experience how it would be to operate a tram enhanced with advanced automatic features. In building the VR simulator, the results of an earlier research study and interviews with conventional tram drivers was used to detect challenging situations faced in tram operators' daily work. For example, the tram diver reacting to sudden movements of pedestrians, cars or trams ahead, maintaining the planned tram schedule, and maintaining the speed and adjusting it according to the varying breaking distance in different weather conditions. The SmartTram simulator system was build based on this contextual understanding of the work of the tram drivers and the VR simulator was aimed at depict the tram driving conditions as realistically as possible. Consequently, the VR simulator offers a high fidelity simulator environment for iterative design and evaluation of the SmartTram project.



Figure 2. A virtual reality based SmartTram simulator. Tram operator's view in an autumn-evening lighting conditions in an urban city environment. The rain and fallen leaves on the tramlines affect acceleration and stopping distance of the tram

The SmartTram simulator system includes 3D models of a tram and a city, road event visualizations (other trams, cars, pedestrians, etc.), different weather conditions (e.g., summer, autumn and winter specific circumstances such as high or low intensity of sunlight and slippery tramlines) and a new SmartTram visual-assistance system. The SmartTram visual-assistance system is based on the three core components: 1) precise model of the tramlines for the current route (e.g., all the turns and altitude changes),

2) tram localization on the route (i.e., GPS/ sensors on the track), and 3) self-awareness system similar to the ones used in the self-driving cars (including sensors for obstacle detection, braking performance calculations, environment conditions monitoring etc.). For example, related to the driving situation in autumn visualized in Figure 2, the SmartTram visual-assistance system is aware of the temperature, humidity and the tramline conditions (e.g., wet). The SmartTram system also uses computer vision to detect presence of leaves on the rails and by using above information, the SmartTram concludes that the rails are very slippery. Depending on the set level of automation, the SmartTram may notify or if needed even transfer the full control to the tram driver to handle the difficult driving situation. Even though the human operator would not have any more the main responsibility of operative driving of the tram, it would still be necessary to have the human in the loop at operating the tram system. In representing information to support human operator's situation awareness about the slippery trail conditions the SmartTram visual-assistance system could, for example change color of the rails and display warning signs on the road surface to indicate the hazard. Similarly, the system may modify corner speed limits or form braking distance estimations, and visualize these for the tram driver. Consequently, the tram driver would then be better aware of the prevailing driving situation and thus be better prepared to take over the control in case needed. Making design decisions on the level of automation and the different visualization, human-system interaction concepts and the different automatic functionalities of the SmartTram may benefit greatly from the possibility to study and explore the qualities of different design solutions (possibly even with the future users) beforehand in a safe simulator environment.

In addition, two hazardous traffic situations (i.e., pedestrians running across the street unexpectedly and a car stopping in front of the tram) were initially introduced in the SmartTram simulator in order to be able to create realistic and challenging test scenarios. Through the scenarios that are based on possible real-life traffic situations in tram driving, it would be also possible to carry out more formal evaluations (e.g., ISV) about the interaction between the human operator and the automatic tram (i.e., the new concept of operation) when the design project is approaching to its end point. A realistic simulator environment is also essential in training the new tram drivers as the imagined role of human operator in operation of SmartTram is substantially different from the role that the dram drivers have today. The build SmartTram simulator system also includes data collection features (e.g., total time, collisions and stopping distance from the tram stop) which are useful when collecting evidence and evaluating the system with the users.

4. CONCLUSIONS AND FUTURE WORK

In the modern smart city environments, many technologically advanced mobility services and solutions are expected to be in place. Thus, increasing levels of automation is also central in developing the public transportation. This paper describes development efforts to introduce automatic tram, that is, the SmartTram. The change from the conventional tram operation to the automatic operation is expected to bring many advances (e.g., efficiency and safety) but also some challenges (e.g., radical change in the human operator's role in driving the tram). Many complex and safety-critical design projects has suffered with lack of proper socio-technical understanding leading costly mistakes in later phases of the system operation. In the SmartTram project, the aim has been to integrate the human and social aspects in the design engineering from the very beginning of the development project. For this purpose, a dedicated HFE program with a V&V plan and a VR based simulator environment was introduced in the project.

In the following phases of our SmartTram project, the most critical HFE activities are to set the user requirements and design an initial concept of operation for the SmartTram. Following the SUC approach the initial goal structure will be formed to support the SmartTram V&V process as well. User studies should be conducted with the SmartTram simulator in order to refine the user requirements and validate the SmartTram operation concept. These activities in the requirements and concept design phase will create a foundation for the later design phases of the development of SmartTram and enables monitoring that the SmartTram project is proceeding favorable direction also from the social and human point of view. Because, in the end, it begins from the smart ´integrative´ design that can make the cities and the transport services of our future smart and sustainable.

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